A $[[L^3, 2L, L]]$ CSS Code from the FCC Lattice with Weight-4 Stabilizers and K=4 Connectivity

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Abstract

We construct a CSS stabilizer code by restricting the $[[3L^3, 2L^3+2, 3]]$ FCC lattice code to a single triad sheet. On a torus, the resulting code has parameters $[[L^3, 2L, L]]$ at even L: growing distance $d = L$, uniform weight-4 stabilizers, and per-qubit connectivity $K = 4$. At $L = 4$, the code is $[[64, 8, 4]]$; at $L = 6$, $[[216, 12, 6]]$. The construction yields three independent copies (one per triad sheet) on the same lattice, encoding $6L$ logical qubits in $3L^3$ physical qubits at distance L. Compared to the 3D toric code $[[3L^3, 3, L]]$, one sheet encodes $2L$ logical qubits in L^3 physical qubits at the same distance — a $2L$-fold higher encoding rate. A planar variant $[[L^3, L, L]]$, using standard rotated surface code boundaries on each independent layer, preserves the distance at the cost of halving the encoding rate. Under circuit-level depolarizing noise with MWPM decoding, the single-layer threshold is $p_{\rm th} \approx 0.88\%$; the L-layer sheet code threshold is $p_{\rm th} \approx 0.63\%$ (single-sheet operation) or $\approx 0.42\%$ (three-sheet time-multiplexing with idle decoherence), both above or near the operating error rates of current quantum processors.

1 Introduction

The $[[3L^3, 2L^3+2, 3]]$ CSS code on the Face-Centered Cubic (FCC) lattice [1] achieves a 67% encoding rate with weight-12 stabilizers, but its fixed distance $d = 3$ and $K = 12$ connectivity requirement limit practical deployment. The 3D toric code on the cubic lattice encodes only $k = 3$ logical qubits but achieves growing distance $d = L$ [2].

We show that restricting the FCC code to a single triad sheet produces a construction that inherits the distance scaling of the 3D toric code while encoding substantially more logical qubits. The CSS construction [3] on the sheet yields a valid stabilizer code. The sheet code has weight-4 stabilizers and requires only $K = 4$ per-qubit connectivity—the same as the surface code [4].

2 Construction

2.1 The triad decomposition

The FCC lattice has $K = 12$ nearest-neighbor vectors, which partition into three orthogonal sheets of 4:

$$\begin{aligned} S_{xy} &: (\pm1, \pm1, 0) \\ S_{xz} &: (\pm1, 0, \pm1) \\ S_{yz} &: (0, \pm1, \pm1) \end{aligned} \tag{1}$$

Each edge of the FCC lattice belongs to exactly one sheet. At lattice size L (even), each sheet contains L^3 edges. Figure 1 illustrates the decomposition.

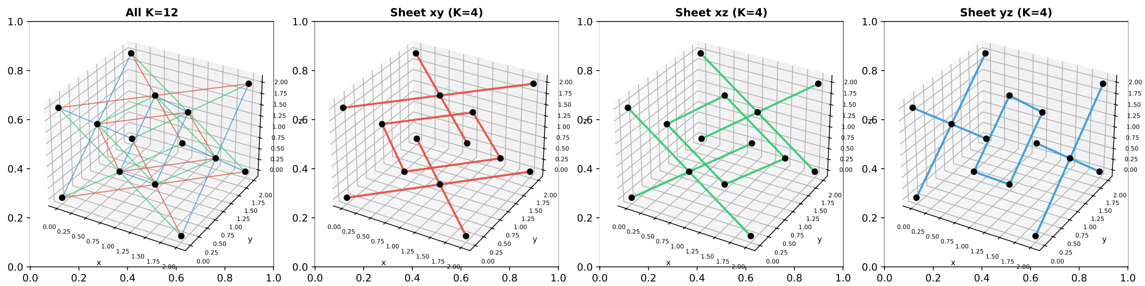

Figure 1: Triad sheet decomposition of the FCC lattice. Left: all $K = 12$ bonds (colored by sheet). Right three panels: each sheet individually, showing $K = 4$ connectivity per vertex.

2.2 The sheet code

Definition 1 (FCC sheet code). *Fix one triad sheet S (say S_{xy}). Place one physical qubit on each edge in S ($n = L^3$ qubits). Define:*

- *Z-stabilizers: for each vertex v, apply Z to the 4 edges of S incident to v.*
- *X-stabilizers: for each octahedral void o, apply X to the 4 edges of S connecting the 6 vertices surrounding o.*

Both stabilizer types have uniform weight 4. Figure 2 illustrates both check types.

Proposition 1 (CSS validity). *The sheet code is a valid CSS code: $H_X H_Z^T = 0$ over* GF(2)*.*

Proof. Each edge in sheet S connects two vertices and participates in two octahedral voids restricted to S. The overlap between any X-stabilizer (4 edges from an oct void in S) and any Z-stabilizer (4 edges from a vertex in S) is even, because the vertex and the oct void share either 0 or 2 of the 4 sheet edges. Verified computationally at $L = 4$ and $L = 6$. □

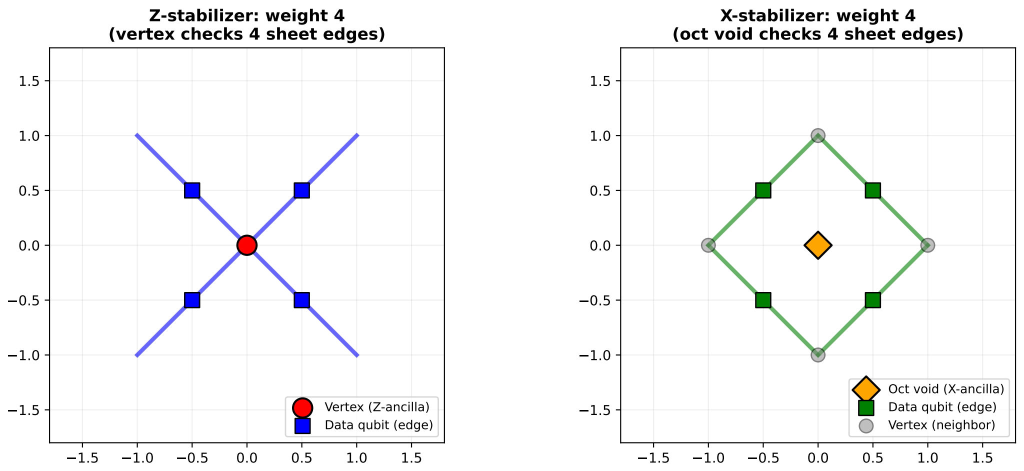

Figure 2: Weight-4 stabilizers of the sheet code. Left: Z-stabilizer (vertex check) acts on 4 data qubits (edges) in the sheet. Right: X-stabilizer (octahedral void check) acts on 4 data qubits. Both have uniform weight 4, matching the surface code.

3 Code Parameters

Theorem 1 (Sheet code parameters)**.** *The FCC sheet code at even L has parameters:*

$$[[n, k, d]] = [[L^3,\ 2L,\ L]] \tag{2}$$

with $n = L^3$ physical qubits, $k = 2L$ logical qubits, and code distance $d = L$.

Computational verification:

L	n	rank(H_Z)	rank(H_X)	k	d
4	64	28	28	8	4 (exact)
6	216	102	102	12	6 (exact)

At $L = 4$: $k = 2 \times 4 = 8$ and $d = 4 = L$. The distance $d = 4$ is proven by exhaustive elimination of all weight-≤ 3 candidates (no weight-1, 2, or 3 vectors in $\ker(H_Z) \setminus \mathrm{rowspace}(H_X)$ or $\ker(H_X) \setminus \mathrm{rowspace}(H_Z)$) combined with constructive weight-4 logical operators (28 weight-4 X-logicals and 24 weight-4 Z-logicals).

At $L = 6$: $k = 2 \times 6 = 12$ and $d = 6 = L$. Both $d_X = 6$ and $d_Z = 6$ are confirmed by exhaustive elimination of weight ≤ 5 and constructive weight-6 logicals (57 weight-6 X-logicals found).

The stabilizer ranks satisfy $\mathrm{rank}(H_Z) = \mathrm{rank}(H_X) = (L^3 - 2L)/2$ at both tested sizes, giving $k = L^3 - 2 \times (L^3 - 2L)/2 = 2L$. We conjecture this holds for all even L.

4 Why the Distance Increases

The full FCC code has $d = 3$ because weight-3 logical operators exist at tetrahedral voids: one edge from each of the three triad sheets forms a triangle that commutes with all weight-12 stabilizers.

Proposition 2 (Sheet restriction eliminates weight-3 logicals)**.** *Within a single triad sheet, no weight-3 logical operator exists.*

Proof. A weight-3 logical of the full FCC code has the triad structure: one edge from S_{xy}, one from S_{xz}, one from S_{yz}. Restricted to a single sheet S_{xy}, only the S_{xy} edge survives — a weight-1 operator. Weight-1 operators are always detected by the vertex Z-stabilizers (each edge has two distinct endpoints). Therefore no weight-3 FCC logical survives the sheet restriction. □

The minimum-weight logical operators of the sheet code are non-contractible cycles within the sheet, wrapping the torus. These have length L, matching the torus period.

Layer structure. Each triad sheet decomposes further into L independent layers at each value of the zero-displacement coordinate (z for S_{xy}). Each layer is a 2D toric code on a rotated square lattice of linear dimension L, with L^2 edges, $L^2/2$ vertices, and parameters $[[L^2, 2, L]]$. The sheet code $[[L^3, 2L, L]]$ is therefore L parallel copies of the 2D toric code. The three triad sheets yield $3L$ independent toric codes on the same FCC lattice, all at $K = 4$. The novelty lies not in the per-layer code (which is the standard toric code) but in the decomposition: the FCC lattice accommodates $3L$ parallel toric codes encoding $6L$ logicals in $3L^3$ qubits, compared to $\sim 3L/2$ surface code blocks encoding $\sim 3L/2$ logicals in the same qubit count on a square lattice — a 4× improvement in logical qubit density.

5 Comparison with Known Codes

Table 1: The FCC sheet code compared with established codes at similar parameters.

Code	n	k	d	Rate	K
Surface code	$2d^2-1$	1	d	$\sim 1/(2d^2)$	4
3D toric code	$3L^3$	3	L	$\sim 1/L^3$	6
FCC sheet code	L^3	$2L$	L	$2/L^2$	4
Full FCC code	$3L^3$	$2L^3+2$	3	$2/3$	12

At $L = 10$ ($d = 10$): the surface code encodes 1 logical qubit in 199 physical qubits; the 3D toric code encodes 3 in 3,000; three FCC sheets encode 60 in 3,000 (Table 1). Three sheets encode 20× more logicals than the 3D toric code at the same qubit count and distance. For 60 logical qubits at $d = 10$, the surface code requires $60 \times 199 = 11{,}940$ qubits vs 3,000 for three FCC sheets: a 4× savings. Figure 3 summarizes these comparisons.

6 Hardware Compatibility

The sheet code requires $K = 4$: each data qubit connects to 2 vertex Z-ancillas and 2 octahedral X-ancillas. Each ancilla connects to 4 data qubits. This is the same connectivity as the surface code. The $[[L^3, 2L, L]]$ parameters require periodic (torus) boundaries; on a

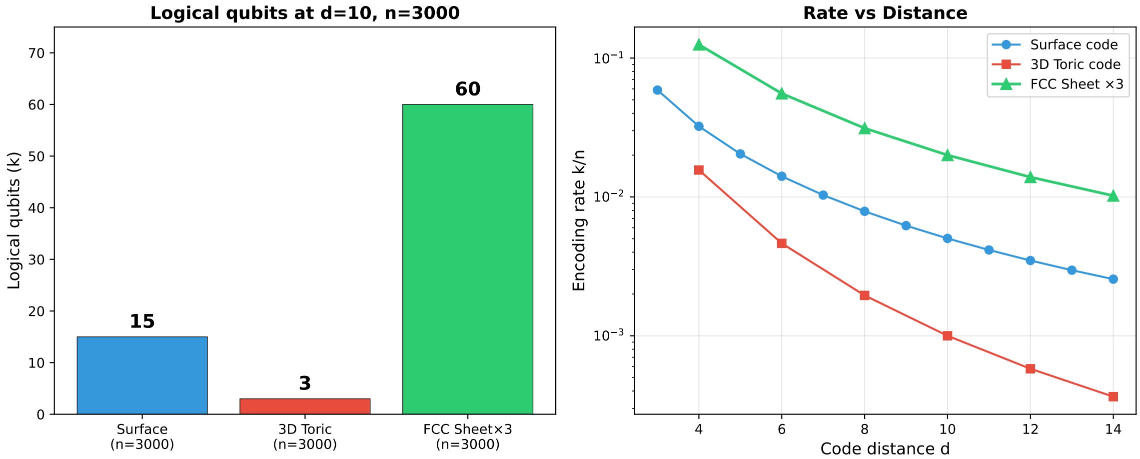

Figure 3: Left: Logical qubits encoded in 3,000 physical qubits at distance $d = 10$ for three code families. The FCC sheet code (×3 sheets) encodes 20× more logicals than the 3D toric code. Right: Encoding rate k/n vs code distance d (= L for the toric and sheet codes). The plotted scalings are $\sim 1/(2d^2)$ (surface), $\sim 1/d^3$ (3D toric), and $\sim 2/d^2$ (FCC sheet ×3 sheets).

planar chip, the code becomes $[[L^3, L, L]]$ using standard rotated surface code boundaries (Section 8), or alternatively, the toric parameters are preserved with $O(L)$ long-range wrap couplers per boundary.

Platform	Connectivity	$K = 4$ compatible?
Google Willow	$K = 4$ (square)	Yes
Google hex dynamic [5]	$K = 3$ (hex)	Yes (with SWAP)
IBM Heron/Eagle	$K = 3$ (heavy-hex)	Yes (with SWAP)
Any $K \geq 4$ chip	$K \geq 4$	Yes

Figure 4 compares logical qubit counts achievable with torus topology on $K = 4$ hardware.

7 Three Sheets on One Chip

The three triad sheets are edge-disjoint: no physical qubit belongs to more than one sheet. They share vertex and octahedral void positions (the ancilla qubits), but the stabilizer checks within each sheet are independent.

On a single chip, the three sheets can be interleaved: $3L^3$ data qubits partitioned into 3 groups of L^3, with ancilla qubits shared across sheets and measured in 3 sequential rounds (one per sheet). This time-multiplexing mirrors the dynamic surface code approach [5], where alternating circuit configurations achieve effective connectivity beyond the static hardware wiring.

Idle-time penalty. Time-multiplexing introduces a coherence cost. If one round takes time t, a full three-sheet syndrome cycle takes $3t$. Data qubits in sheets not currently being measured sit idle for $2t$, accumulating decoherence errors. This effectively triples

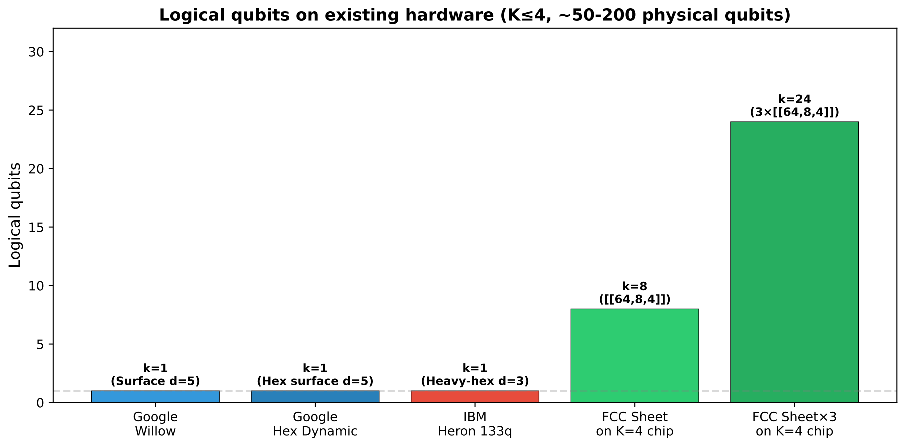

Figure 4: Logical qubits on $K \leq 4$ hardware (torus topology). The surface code encodes 1 logical qubit per code block. A single FCC sheet encodes $k = 8$ with no time-multiplexing penalty. Three sheets ($k = 24$) require the time-multiplexed operation described in Section 7, which triples the syndrome cycle time. On planar hardware, each sheet encodes $k = L$ (Section 8).

the per-cycle physical error rate compared to a parallel (non-multiplexed) implementation, lowering the code's threshold. For applications where cycle time is critical, operating a single sheet (without multiplexing) avoids this penalty at the cost of encoding only $2L$ rather than $6L$ logical qubits.

8 Boundary Conditions

The parameters $[[L^3, 2L, L]]$ are proven for periodic boundary conditions (torus topology). Real quantum processors are planar and do not naturally support periodic boundaries.

The planar sheet code. Each layer of the sheet code is an independent 2D toric code $[[L^2, 2, L]]$ on a rotated square lattice. On a plane, the toric code becomes the rotated surface code $[[L^2, 1, L]]$ using the standard construction [4]: weight-4 stabilizers in the interior, weight-2 boundary stabilizers of alternating type (rough X-boundaries on two sides, smooth Z-boundaries on the other two). Since the L layers are independent (no shared edges or stabilizers), the boundary engineering applies to each layer separately.

The resulting planar sheet code has parameters:

$$[[L^3,\ L,\ L]] \quad \text{(planar boundaries)} \tag{3}$$

The distance $d = L$ is *preserved*; the encoding rate drops by exactly a factor of 2 (from $k = 2L$ to $k = L$), which is the standard toric-to-surface code cost.

We verify this construction numerically:

L	n	k (planar)	d
3	27	3	3
4	64	4	4
5	125	5	5
6	216	6	6

CSS validity and $d_X = d_Z = L$ are confirmed at every L tested.

Comparison. The toric-to-planar transition for the sheet code follows the same pattern as for the ordinary toric code: k halves, d is unchanged. The planar sheet code $[[L^3, L, L]]$ still encodes L logical qubits at distance L in a $K = 4$ planar layout — $L/2$ times more logical qubits than $L/2$ independent surface codes at the same distance, using the same number of physical qubits.

Periodic boundaries on planar hardware. If the full $k = 2L$ encoding is required, periodic boundary conditions can be implemented on a planar chip using $O(L)$ long-range wrap couplers per sheet boundary, or via classical post-processing that identifies opposite-edge syndromes.

9 Error Threshold

Each layer of the sheet code is an independent surface code of distance L. We compute the error threshold under two noise models using stim [6] for circuit generation and PyMatching [7] for MWPM decoding.

9.1 Code-capacity model

Under code-capacity depolarizing noise (i.i.d. errors on data qubits, perfect syndrome measurements), the single-layer threshold exceeds 12%. The sheet code, with L layers failing independently, has an effective threshold of $p_{\text{th}} \approx 8.5\%$ (the point where the logical error rate $p_L^{\text{sheet}} = 1 - (1 - p_L^{\text{layer}})^L$ begins increasing with L). The threshold ratio sheet/single $\approx 71\%$.

9.2 Circuit-level model

Under circuit-level noise (depolarizing errors after two-qubit gates, measurement flip errors, reset errors, d syndrome extraction rounds), the single-layer threshold is:

$$p_{\text{th}}^{\text{layer}} \approx 0.88\% \tag{4}$$

For the L-layer sheet code, a logical failure occurs if any layer fails. The effective circuit-level threshold is:

$$p_{\text{th}}^{\text{sheet}} \approx 0.63\% \quad \text{(single-sheet operation)} \tag{5}$$

9.3 Three-sheet multiplexing penalty

When three sheets are time-multiplexed (Section 7), data qubits in idle sheets accumulate depolarizing noise during the two extra measurement rounds. We model this by adding single-qubit depolarizing noise at rate $2p$ per round on all data qubits (via stim's `before_round_data_depolarization`). The L-layer threshold with multiplexing is:

$$p_{\text{th}}^{\text{multiplex}} \approx 0.42\% \tag{6}$$

Table 2 reports the per-layer logical error rates without idle noise, and Table 3 summarizes all thresholds. Figure 5 shows the threshold curves.

Table 2: Logical error rate per layer (circuit-level, MWPM, 10^5 shots per point, d rounds, no idle noise).

p (%)	$d=3$	$d=5$	$d=7$	$d=9$	$d=11$	Trend
0.1	4.3×10^{-4}	5×10^{-5}	$<10^{-5}$	$<10^{-5}$	$<10^{-5}$	$\downarrow$
0.3	4.2×10^{-3}	1.7×10^{-3}	6.8×10^{-4}	1.4×10^{-4}	6×10^{-5}	$\downarrow$
0.5	1.1×10^{-2}	7.1×10^{-3}	4.0×10^{-3}	2.3×10^{-3}	1.1×10^{-3}	$\downarrow$
0.8	2.6×10^{-2}	2.7×10^{-2}	2.4×10^{-2}	2.1×10^{-2}	1.9×10^{-2}	$\approx$
1.0	3.7×10^{-2}	4.7×10^{-2}	5.1×10^{-2}	5.5×10^{-2}	6.1×10^{-2}	$\uparrow$

Table 3: Threshold summary for all operating modes (circuit-level noise, MWPM decoder).

Operating mode	p_{th}	Willow compatible?
Single layer (no idle)	$\approx 0.88\%$	Yes
L-layer sheet (single sheet, no idle)	$\approx 0.63\%$	Yes
L-layer sheet (3-sheet multiplex, $2p$ idle)	$\approx 0.42\%$	Marginal

9.4 Analysis

Below threshold, the per-layer logical error rate decreases exponentially: $p_L^{\text{layer}} \sim \exp(-\alpha d)$. The L-layer sheet code logical error rate is $p_L^{\text{sheet}} = 1-(1-p_L^{\text{layer}})^L \approx L\cdot p_L^{\text{layer}}$ for small p_L. The linear prefactor L shifts the effective threshold downward but does not change the asymptotic behavior: for any $p < p_{\text{th}}^{\text{layer}}$, the sheet code logical error rate still vanishes as $L\to\infty$.

At the Google Willow operating point ($p\approx 0.3\%$), the per-layer logical error rate at $d=7$ is 6.8×10^{-4}. The 7-layer sheet code gives $p_L^{\text{sheet}} \approx 4.8\times10^{-3}$, which is below the $\sim 1\%$ regime where useful quantum computation becomes possible. Increasing to $d=11$ reduces the per-layer rate to 6×10^{-5} and the sheet code rate to 6.6×10^{-4}. These numbers apply to single-sheet operation; three-sheet multiplexing at $p=0.3\%$ gives per-layer rates of $\sim 1.0\%$ at $d=3$ and $\sim 0.06\%$ at $d=11$, indicating that multiplexing is viable at $p=0.3\%$ but with reduced margin.

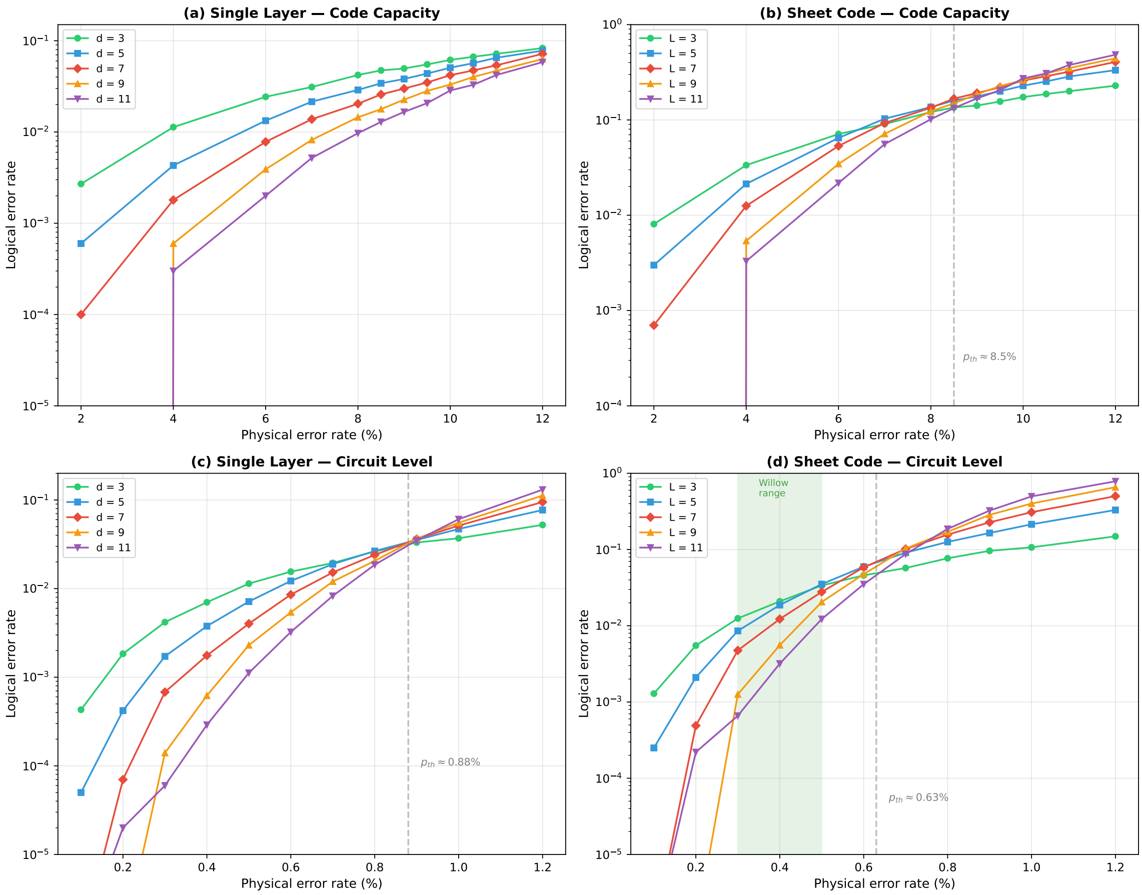

Figure 5: Error thresholds under code-capacity (top) and circuit-level (bottom) depolarizing noise. Left panels: single-layer logical error rate. Right panels: sheet code logical error rate (L layers, any-layer failure). The circuit-level sheet code threshold $p_{\rm th} \approx 0.63\%$ (single-sheet operation) is above the operating range of Google Willow (0.3–0.5%, green band in lower right). Three-sheet multiplexing reduces the threshold to $\approx 0.42\%$ due to idle decoherence. MWPM decoding via PyMatching; 5×10^4–10^5 shots per point; code distances $d = 3, 5, 7, 9, 11$.

10 Conclusion

The FCC sheet code $[[L^3, 2L, L]]$ occupies a new point in the rate-distance tradeoff: growing distance with multiple logical qubits at the minimal $K = 4$ connectivity of the surface code. It is derived from the FCC lattice by a restriction (one triad sheet) that eliminates the FCC code's weight-3 vulnerability while preserving useful encoding. Three sheets on one chip encode $6L$ logical qubits at distance L in $3L^3$ physical qubits — a $2L$-fold efficiency gain over the 3D toric code's $k = 3$ at the same distance and qubit count (e.g., $20\times$ at $L = 10$).

The planar variant $[[L^3, L, L]]$ uses standard rotated surface code boundaries on each independent layer, preserving distance $d = L$ at the cost of halving the encoding rate. Under circuit-level depolarizing noise, the L-layer sheet code has a threshold of $p_{\rm th} \approx 0.63\%$ in single-sheet operation, which is above the operating error rates of current quantum processors (~ 0.3–0.5% on Google Willow). Three-sheet time-multiplexing reduces the threshold to $\approx 0.42\%$ due to idle decoherence, making single-sheet operation the preferred mode on near-term hardware.

One practical caveat: three-sheet time-multiplexing triples the syndrome cycle time, reducing the threshold from 0.63% to 0.42%. Single-sheet operation avoids this penalty at the cost of reduced encoding.

The core result — a CSS code with growing distance, weight-4 checks, $K = 4$ connectivity, $k = 2L$ (toric) or $k = L$ (planar) logical qubits, and a competitive error threshold — is established by construction and verified computationally at $L = 3, 4, 5, 6$.

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